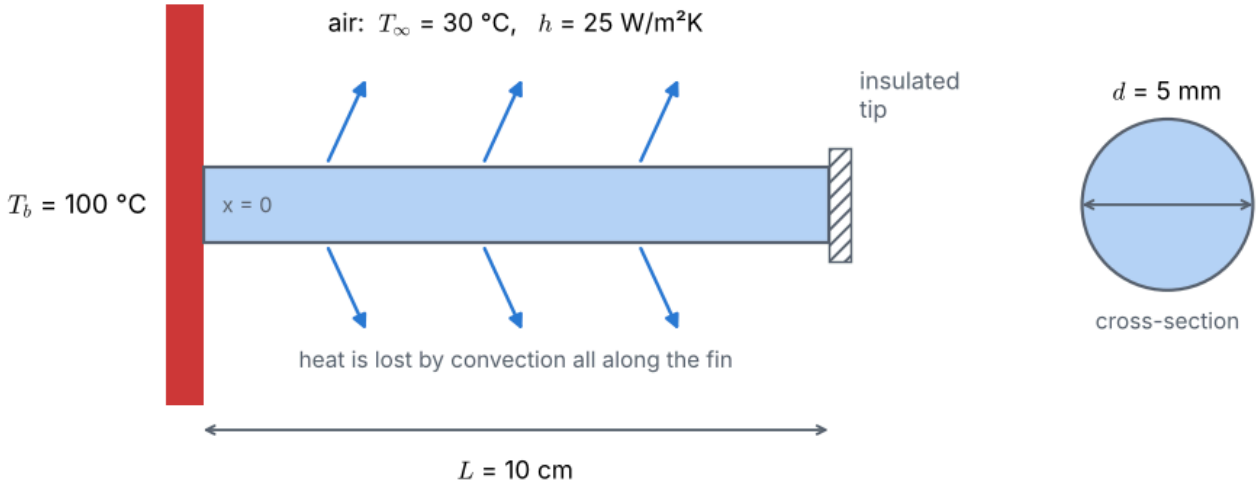
PROBLEM 1 · PROBLEM STATEMENT

Steady temperature distribution in an aluminium pin fin

One-dimensional steady heat conduction with convection, solved with the Finite Difference Method

The problem

> Determine the **steady-state temperature distribution** within an **aluminium fin** using the **Finite Difference Method**. Assume the base temperature of the fin is **100 °C** and the tip of the fin is **insulated**. Initially the fin is at a temperature of **30 °C**.



A cylindrical aluminium pin fin attached to a wall at T_b. Heat is conducted along the fin and convected from its surface to the surrounding air; the tip is insulated.

Data

The statement fixes the base temperature, the tip condition and the initial temperature. The remaining data were not given and are taken as **typical values** (you may change them in the notebook):

Quantity	Symbol	Value
fin length	L	10 cm
fin diameter (circular pin)	d	5 mm
thermal conductivity of aluminium	k	237 W/(m·K)
convection heat transfer coefficient	h	25 W/(m²·K)
temperature of the surrounding air	T_∞	30 °C
base temperature	T_b	100 °C
initial temperature of the fin	T_{init}	30 °C

Assumptions

- Steady state: the temperature no longer changes with time.
- One-dimensional conduction along the fin axis x: the fin is thin, so the temperature is uniform over each cross-section.
- Constant thermal conductivity k and convection coefficient h along the whole fin.
- Radiation is neglected.
- The tip of the fin is insulated (no heat leaves through it).

Tasks

1. **Physics.** Write an energy balance for a thin slice of the fin and derive the governing equation for $T(x)$. State the two boundary conditions.
2. **Grid.** Divide the fin into $N = 10$ equal intervals and number the nodes from the base ($i = 0$) to the tip ($i = N$).
3. **Discretization.** Replace the second derivative by a central finite difference and write one algebraic equation for every node. Use a *ghost node* to apply the insulated-tip condition.
4. **Solution.** Solve the system of equations
   a. with the **Gauss–Seidel** iterative method, starting from the initial temperature of 30 °C;
   b. with the **Thomas algorithm** (direct solver for tridiagonal systems).
5. **Results.** Plot the temperature distribution and find the **tip temperature** and the **heat lost by the fin**.
6. **Verification.** Compare with the exact solution and repeat the calculation on finer grids. How does the error change when Δx is halved?

What you will learn

- How an energy balance leads to a differential equation, and how boundary conditions complete it.
- How the finite difference method turns a differential equation into a system of algebraic equations.
- How to apply a derivative (Neumann) boundary condition with a ghost node.
- The difference between iterative and direct solvers.
- How to verify a numerical solution with an exact solution and a grid-refinement study.

> **Files for this problem:** `02_solution.pdf` explains the solution step by step. `03_fdm_solver.ipynb` contains the complete Python code (plain Python, no numerical libraries), with explanations and plots, to run cell by cell.